

Justin Finkel
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<https://justinfocus12.github.io>

Education

- 2017-2022 **Ph.D.**, Computational and Applied Mathematics, University of Chicago
Thesis topic: Atmospheric extremes through the lens of transition path theory
Advisor: Jonathan Weare (NYU)
- 2013-2017 **B.A.**, Physics and Mathematics, *Magna Cum Laude*, Washington University in St. Louis
Thesis topic: Changing World Temperature Statistics
Advisor: Jonathan Katz

Employment

- 2025– **Postdoctoral Fellow**, Data Science Institute and Department of Geophysical Sciences, University of Chicago (Advisors: Pedram Hassanzadeh, Dorian Abbot)
- 2022-2025 **Postdoctoral Associate**, Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology (Advisor: Paul O’Gorman)
- 2019 **Summer intern**, Los Alamos National Laboratory
- 2017-2020 **Online tutor**, Varsity Tutors
- 2013-2017 **Academic mentor**, Cornerstone Learning Center, Washington University in St. Louis
- 2016 **Summer intern**, MIT Lincoln Laboratory
- 2015 **Research assistant**, Remote Sensing Laboratory, Department of Earth and Planetary Sciences, Washington University in St. Louis (Advisor: Raymond Arvidson)
- 2014 **Research assistant**, Department of Physics and McDonnell Center for the Space Sciences, Washington University in St. Louis (Advisors: Christine Floss and Ernst Zinner)

Publications

Submitted

1. **Finkel**, J. & O’Gorman, P. A. Boosting ensembles for statistics of tails at conditionally optimal advance split times. arXiv: [2507.22310 \[physics.ao-ph\]](#) (2025).
2. **Finkel**, J. & O’Gorman, P. A. Rare event sampling for moving targets: extremes of temperature and daily precipitation in a general circulation model. arXiv: [2508.13120 \[physics.ao-ph\]](#) (2025).
4. Packman, S., **Finkel**, J., Abbot, D. S. & Tziperman, E. Using a rare event sampling technique to quantify extreme El Niño event statistics. arXiv: [2512.23038 \[physics.ao-ph\]](#) (2025).
5. Seviour, W. J. M., **Finkel**, J., Rupp, P., *et al.* Forecast-based attribution of the role of stratospheric variability in weather extremes (2025).

Published

6. **Finkel**, J. & O’Gorman, P. A. Bringing Statistics to Storylines: Rare Event Sampling for Sudden, Transient Extreme Events. *Journal of Advances in Modeling Earth Systems* **16**, e2024MS004264. eprint: <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2024MS004264> (2024).
10. Zhang, H., **Finkel**, J., Abbot, D. S., Gerber, E. P. & Weare, J. Using Explainable AI and Transfer Learning to Understand and Predict the Maintenance of Atlantic Blocking With Limited Observational Data. *Journal of Geophysical Research: Machine Learning and Computation* **1**, e2024JH000243. eprint: <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2024JH000243> (2024).
11. **Finkel**, J., Gerber, E. P., Abbot, D. S. & Weare, J. Revealing the Statistics of Extreme Events Hidden in Short Weather Forecast Data. *AGU Advances* **4**, e2023AV000881. eprint: <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2023AV000881> (2023).
16. Strahan, J., **Finkel**, J., Dinner, A. R. & Weare, J. Predicting rare events using neural networks and short-trajectory data. *Journal of Computational Physics* **488**, 112152. ISSN: 0021-9991. <https://www.sciencedirect.com/science/article/pii/S0021999123002474> (2023).
17. **Finkel**, J., Webber, R. J., Gerber, E. P., Abbot, D. S. & Weare, J. Data-driven transition path analysis yields a statistical understanding of sudden stratospheric warming events in an idealized model. *Journal of the Atmospheric Sciences*. <https://journals.ametsoc.org/view/journals/atsc/aop/JAS-D-21-0213.1/JAS-D-21-0213.1.xml> (2022).
20. **Finkel**, J., Webber, R. J., Gerber, E. P., Abbot, D. S. & Weare, J. Learning Forecasts of Rare Stratospheric Transitions from Short Simulations. *Monthly Weather Review* **149**, 3647–3669. <https://journals.ametsoc.org/view/journals/mwre/149/11/MWR-D-21-0024.1.xml> (2021).
24. **Justin Finkel**, Abbot, D. S. & Weare, J. Path Properties of Atmospheric Transitions: Illustration with a Low-Order Sudden Stratospheric Warming Model. *Journal of the Atmospheric Sciences* **77**, 2327–2347. <https://journals.ametsoc.org/view/journals/atsc/77/7/jasD190278.xml> (2020).
25. Popović, P., **Finkel**, J., Silber, M. C. & Abbot, D. S. Snow Topography on Undeformed Arctic Sea Ice Captured by an Idealized “Snow Dune” Model. *Journal of Geophysical Research: Oceans* **125**, e2019JC016034. eprint: <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2019JC016034> (2020).
26. **Finkel**, J. M. & Katz, J. I. Changing world extreme temperature statistics. *International Journal of Climatology* **38**, 2613–2617. eprint: <https://rmets.onlinelibrary.wiley.com/doi/pdf/10.1002/joc.5342> (2018).
27. **Finkel**, J. M. & Katz, J. I. Changing US extreme temperature statistics. *International Journal of Climatology* **37**, 4749–4755. eprint: <https://rmets.onlinelibrary.wiley.com/doi/pdf/10.1002/joc.5115> (2017).
28. Kreisch, C., O’Sullivan, J., Arvidson, R., Politte, D., He, L., Stein, N., **J. Finkel**, Guinness, E., Wolff, M. & Lapôtre, M. Regularization of Mars Reconnaissance Orbiter CRISM along-track oversampled hyperspectral imaging observations of Mars. *Icarus* **282**, 136–151. ISSN: 0019-1035. <https://www.sciencedirect.com/science/article/pii/S0019103516301579> (2017).

29. **Finkel**, J. M., Canel-Katz, L. M. & Katz, J. I. Decreasing US aridity in a warming climate. *International Journal of Climatology* **36**, 1560–1564. eprint: <https://rmets.onlinelibrary.wiley.com/doi/pdf/10.1002/joc.4421> (2016).

Awards & Honors

2025	Seed grant from the Institute for Climate and Sustainable Growth, University of Chicago, for the project AI-supercharged Climate Extreme Event Risk Assessment
2018-2022	Department of Energy Computational Science Graduate Fellowship
2017	Nishi Luthra award for “Outstanding students in the Physics department and the Philosophy department”
2017	<i>Sigma Pi Sigma</i> inductee (physics honor society)
2015	Academic Mentor of the Year (for physics mentoring)

Teaching & Mentorship

2024-2025	Co-advised an undergraduate thesis by Sarah Packman, Harvard University: “Using a rare event sampling technique to quantify extreme El Nino event statistics”
2024	Supervised an MIT Summer Research Project by Jesus Lopez, Texas AM San Antonio: “Spatiotemporal Dynamics and Rare Event Sampling of Local Extremes in Passive Tracer Distributions using Simplified Models”
2020	Developed and taught (together with faculty Mary Silber) an intensive online short course in linear algebra for incoming chemistry Ph.D. students at UChicago.
2017-2020	Online tutoring in math, physics, and programming at high school, undergraduate, and graduate level for students of widely varied backgrounds.
2014-2017	Worked with and led a team of mentors for introductory physics sequence at Washington University in St. Louis.

Presentations

1. University of Chicago, computational and applied math seminar October 2025. Invited talk.
2. University of Texas at Austin, Oden Institute, Rising Stars in Scientific Computing, April 2025. Contributed talk.
3. University of Colorado, Boulder applied math seminar, December 2024. Invited talk.
4. Cornell Earth and Atmospheric Science seminar, November 2024. Invited talk.
5. Yale Atmospheric, Ocean, and Climate Dynamics seminar, October 2024. Invited talk.
6. Cross-VESRI (Schmidt Sciences) convening, July 2024, Cambridge, UK. Contributed talk.
7. Atmospheric and Oceanic Fluid Dynamics meeting, June 2024. Poster.
8. University of Chicago Institute for Mathematical and Statistical Innovation, September-October 2022. Invited talks.

9. American Physical Society March Meeting, 2022. Contributed talk. March 14, 2022.
10. Center for Atmosphere Ocean Science, New York University Courant Institute seminar. Joint with Jonathan Weare. October 13, 2021
11. University of Bristol Climate Dynamics seminar. Joint with Dorian Abbot. May 26, 2021
12. European Geophysical Union General Assembly, 2021. Contributed talk. April 28, 2021
13. SIAM conference on dynamical systems, 2021. Contributed talk. May 23, 2021
14. American Physical Society March Meeting, 2021. Contributed talk. March 18, 2021
15. Courant Institute of Mathematical Sciences student seminar, 2021. March 12, 2021
16. SIAM conference on dynamical systems, 2019. Poster. May 22, 2019
17. American Geophysical Union Fall Meeting, 2018. Contributed talk. December 10, 2018
18. Midstates Consortium for Math and Science, Nov. 5, 2016. Poster. November 5, 2016
19. Washington University Undergraduate Research Symposium. October 2014. Poster.

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